

CSEC Physics January 2007 P1

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1. Which of the following is NOT a fundamental quantity?

(A) Temperature
(B) Heat energy
(C) Length
(D) Time

2. The unit of electrical resistance may be expressed as

(A) $1 \Omega = \text{IV} \cdot \text{A}^{-1}$
(B) $1 \Omega = \text{I A} \cdot \text{V}$
(C) $1 \Omega = \text{IV}^{-1} \cdot \text{A}$
(D) $1 \Omega = \text{IWA}^{-1}$

3. In which of the following groups are the four electric currents listed in order of INCREASING size?

(A) $1 \mu\text{A}$, $1 \times 10^3 \text{ A}$, 0.01 A , 100 mA
(B) $1 \mu\text{A}$, 100 mA , 0.01 A , $1 \times 10^3 \text{ A}$
(C) $1 \times 10^3 \text{ A}$, $1 \mu\text{A}$, 100 mA , 0.01 A
(D) $1 \mu\text{A}$, 0.01 A , $1 \times 10^3 \text{ A}$, 100 mA

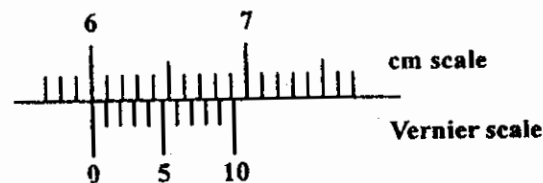
4. A student was asked to express the experimental result for the density of mercury in standard form. Which of the following is the correct way of stating the result?

(A) $0.1362 \times 10^3 \text{ kgm}^{-3}$
(B) $1.362 \times 10^4 \text{ kgm}^{-3}$
(C) $13.62 \times 10^3 \text{ kgm}^{-3}$
(D) $136.2 \times 10^2 \text{ kgm}^{-3}$

5. Which of the following is suitable for measuring the diameter of a human hair?

(A) Tape measure
(B) Vernier caliper
(C) Micrometer screw gauge
(D) Metre rule

Item 6 refers to the following diagram.



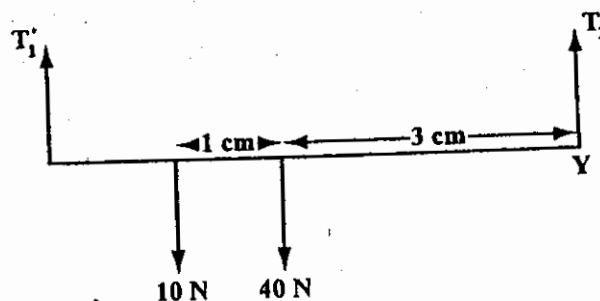
6. The diagram above indicates a vernier caliper measurement of

(A) 6.00 cm
(B) 6.02 cm
(C) 6.10 cm
(D) 6.11 cm

7. The force which keeps the moon in orbit is

(A) magnetic
(B) centrifugal
(C) reactive
(D) gravitational

Item 8 refers to the diagram below.

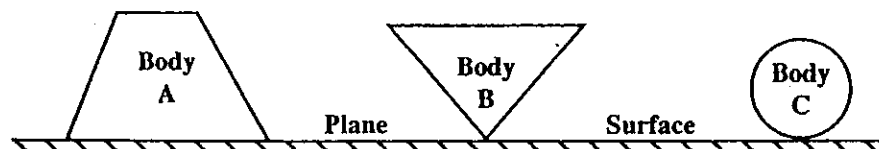


8. A uniform rod XY of weight 40 N is held in equilibrium, horizontally, as shown in the diagram above. Which of the following is true?

(A) $T_1 < T_2$
(B) $T_1 \leq T_2$
(C) $T_1 = T_2$
(D) $T_1 > T_2$

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9. In relation to the diagrams below, which of the following statements is/are correct?



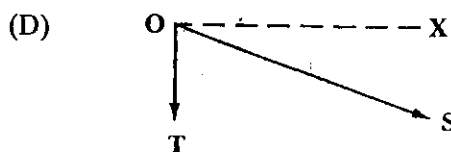
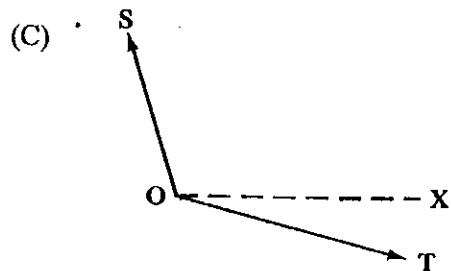
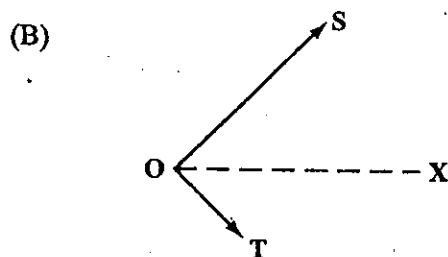
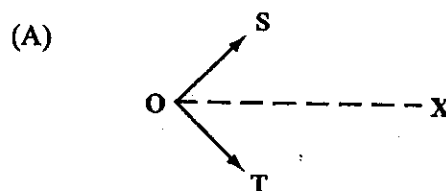
- I. Body A is in neutral equilibrium.
- II. Body B is in unstable equilibrium.
- III. Body C is in stable equilibrium.

- (A) II only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

10. What is the unit of the moment of a force?

- (A) N
- (B) N m
- (C) N m^{-1}
- (D) N m^{-2}

11. The diagrams below, drawn to scale, represent two forces S and T acting at O. In which of the following is the resultant in the direction, OX?



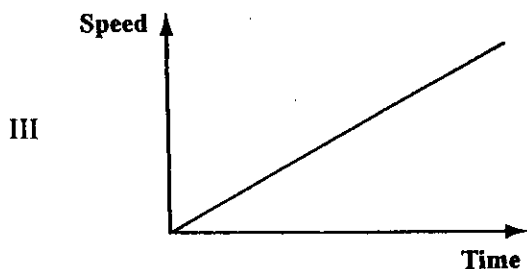
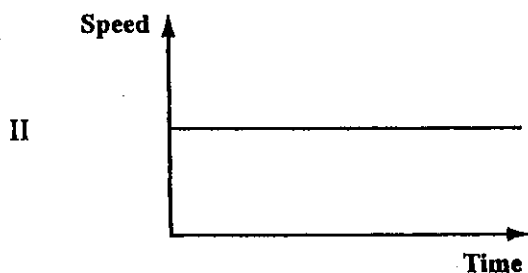
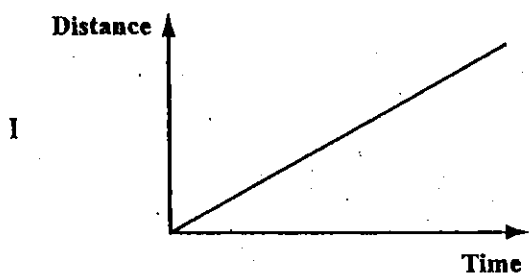
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12. Which of the following could be used to find the speed of an object?

- (A) $\frac{\text{Change in velocity}}{\text{Time taken}}$
- (B) $\frac{\text{Change in displacement}}{\text{Time taken}}$
- (C) $\frac{\text{Distance travelled}}{\text{Time taken}}$
- (D) Distance travelled x Time taken

13. Which of the following graphs represents a constant speed?



- (A) I only
- (B) II only
- (C) I and II only
- (D) I and III only

14. Which of the following is the unit of momentum?

- (A) m s^{-1}
- (B) kg m s^{-1}
- (C) kg m s^{-2}
- (D) N s

15. The mass of a rifle is 100 times the mass of its bullet. If the recoil speed of the rifle is v , then the speed of the bullet is

- (A) $100v^2$
- (B) $\frac{v^2}{100}$
- (C) $100v$
- (D) $\frac{v}{100}$

16. Which of the following implements is/are designed to take advantage of a large moment provided by a relatively small force?

- I. Claw hammer
- II. Crowbar
- III. Pair of tweezers
- IV. Pair of wire cutters

- (A) III only
- (B) I and IV only
- (C) I, II and III only
- (D) I, II and IV only

17. A solid object when placed in a beaker of water just sinks to the bottom. It could be made to float by doing which of the following?

- I. Cutting off part of the object
- II. Dissolving a lot of salt in the water
- III. Carrying the beaker up to the top of a high mountain

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only

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18. The kinetic energy of a body of mass, m , and velocity, v , is

- (A) $\frac{m}{v}$
(B) mv
(C) $\frac{mv^2}{2}$
(D) mv^2

19. A glass marble, X, moving with a speed of 6 m s^{-1} , collides 'headon' with an identical stationary glass marble, Y. What is the velocity of Y after collision assuming that X is brought to rest?

- (A) 0 m s^{-1}
(B) 3 m s^{-1}
(C) 6 m s^{-1}
(D) 12 m s^{-1}

20. A thermocouple may be used as a thermometer. Which of the following properties is utilised in a thermocouple?

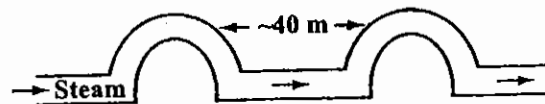
- (A) Expansion of a metal
(B) Expansion of two different metals
(C) Production of electric current at a junction of two wires
(D) Production of electric current in a heated wire

21. Absolute zero is the temperature at which

- (A) air changes to a liquid
(B) a gas would exert no pressure
(C) all substances contract to zero volume
(D) gas molecules move at a constant speed

Items 22 refer to the following information.

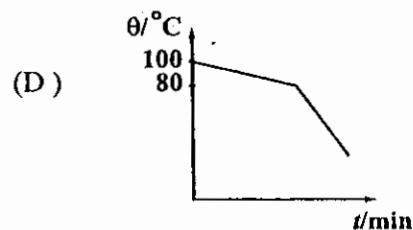
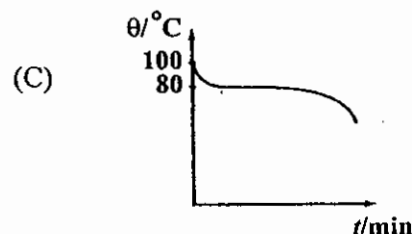
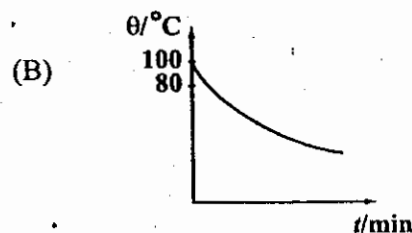
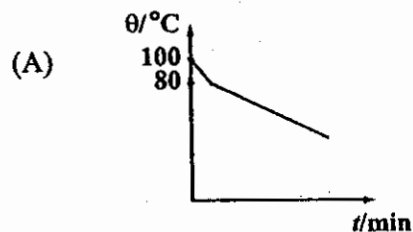
In a hospital, the steam pipes hung from the ceiling in the corridor are not straight but rather have a U bend at intervals of about 40 m, as shown in the diagram below.



22. The purpose of the bends in the pipe is to

- (A) trap any water in the steam
(B) reduce the pressure of the steam
(C) conduct heat away from the pipes
(D) allow for expansion of the pipes

23. Some molten naphthalene at 100°C is allowed to cool down at room temperature. If naphthalene has a melting point of 80°C , which of the following BEST represents the cooling curve?



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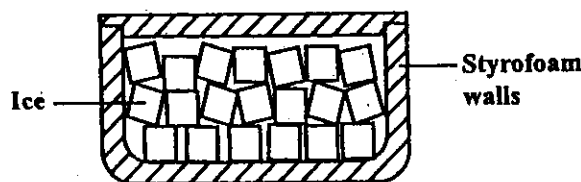
24. The specific latent heat of vaporization of water is the energy required to change 1 kg of water at

- (A) 0°C to steam at 100°C
- (B) 99.9°C to steam at 100.1°C
- (C) 100°C to steam at 100°C
- (D) 0°C to ice at 0°C

25. Which of the following would be responsible for the heating of the Earth by the sun?

- I. Radiation
 - II. Convection
 - III. Conduction
- (A) I only
 - (B) III only
 - (C) I and II only
 - (D) II and III only

Item 26 refers to the following diagram.



A styrofoam "igloo" (see diagram above) for keeping drinks cool was filled with ice in the morning. By evening, all the ice had melted. This is because it

- (A) lost heat through the walls by radiation
- (B) lost heat through the walls by conduction
- (C) gained heat through the walls by radiation
- (D) gained heat through the walls by conduction

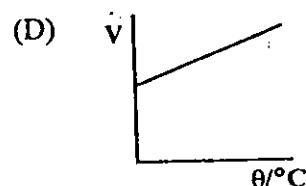
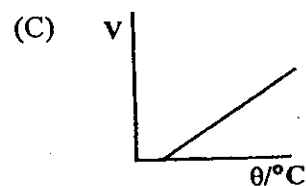
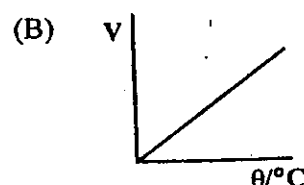
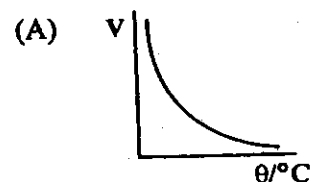
27. Deep water $\longrightarrow$ Shallow water

Plane water waves travel from deep water to shallow water with an angle of incidence of 0° as shown in the diagram above. Which of the following would be true?

- I. The frequency of the waves decreases.
- II. The speed of the waves decreases.
- III. The wavelength of the waves decreases.

- (A) I, II and III
- (B) I and II only
- (C) I and III only
- (D) II and III only

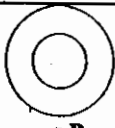
28. Which of the following graphs relates the volume of a gas to its Celsius temperature if the mass of gas and pressure remain constant?

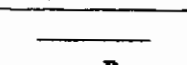


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29. A pencil point is dipped into water at Point P in front of a straight barrier. The REFLECTED wave will look like

(A)  Barrier

(B)  Barrier

(C)  Barrier

(D)  Barrier

30. When a special whistle is blown, dogs can hear the signal but people cannot. Which of the following could be the frequency of the waves emitted by the whistle?

- I. 1 500 Hz
- II. 15 000 Hz
- III. 25 000 Hz

- (A) III only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

31. Compared with X-rays, radiowaves

- (A) have a longer wavelength
- (B) have a higher frequency
- (C) travel faster in air
- (D) pass through a sheet of steel more easily

32. The sharp edges of shadows suggest that light

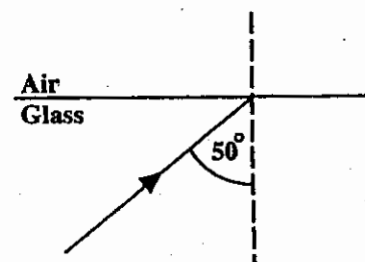
- (A) has a wave nature
- (B) is a form of energy
- (C) travels in straight lines
- (D) travels very quickly

33. Which of the following are true of the image produced by a plane mirror?

- I. It is enlarged.
- II. It is erect.
- III. It is vertical.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

Item 34 refers to the following diagram.



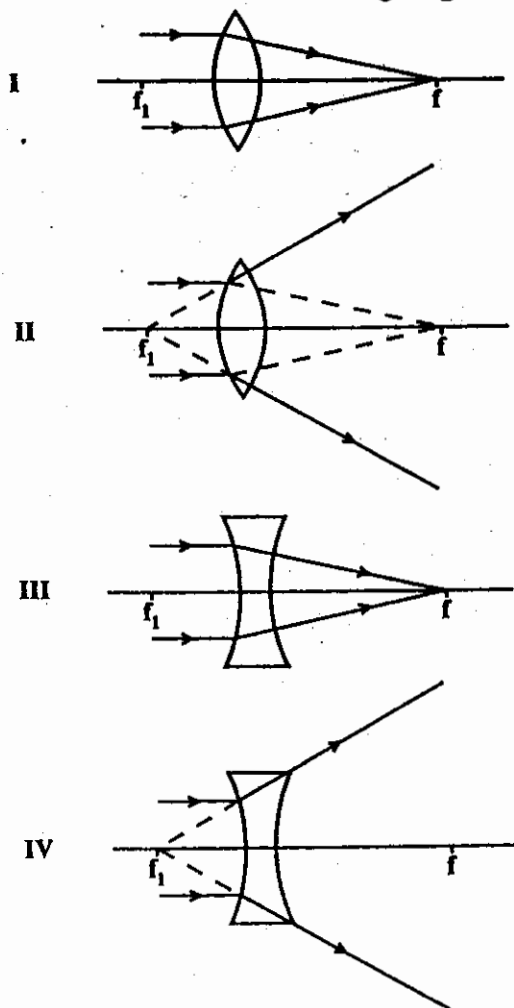
34. The light ray shown in the diagram above is incident at the air-glass interface at an angle of incidence of 50° . If the critical angle for glass is 45° , the light ray will be

- (A) reflected
- (B) refracted
- (C) dispersed
- (D) diffracted

35. For which of the following object distances will a convex lens of focal length 18 cm produce a real image?

- I. 15 cm
 - II. 36 cm
 - III. 54 cm
- (A) I only
(B) I and II only
(C) II and III only
(D) I, II and III

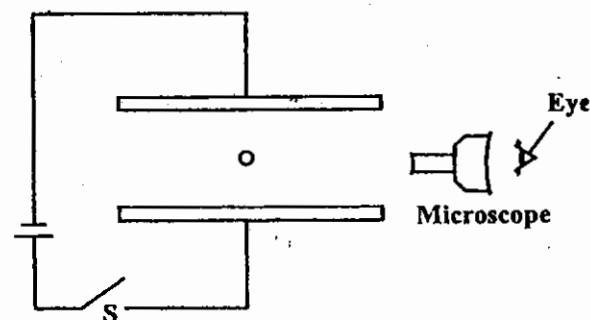
Item 36 refers to the following diagrams.



36. Which of the ray diagrams shown above are correct?

- (A) II and III only
(B) II and IV only
(C) I and III only
(D) I and IV only

Item 37 refers to the following diagram.



37. The diagram above shows a small negatively charged dust particle midway between two plates and observed with a microscope.

Which of the following arrows correctly indicates the path taken by the dust particle when the switch S is closed?

- (A) $\rightarrow$
(B) $\leftarrow$
(C) $\uparrow$
(D) $\downarrow$

38. Which of the following equations is NOT correct?

- (A) $1 \text{ C} = 1 \text{ A} \times 1 \text{ s}$
(B) $1 \text{ V} = 1 \text{ A} \times 1$
(C) $1 \text{ J} = 1 \text{ A} \times 1 \Omega$
(D) $1 \text{ W} = 1 \text{ V} \times 1 \text{ C}$

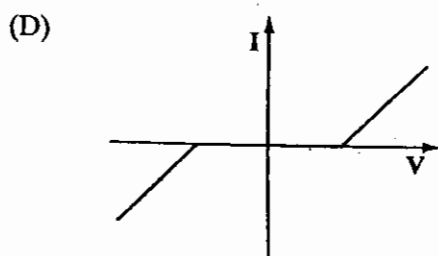
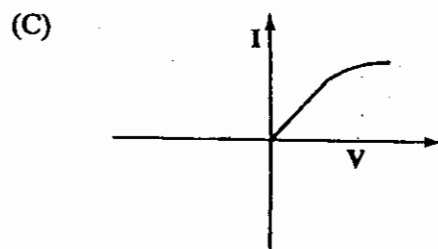
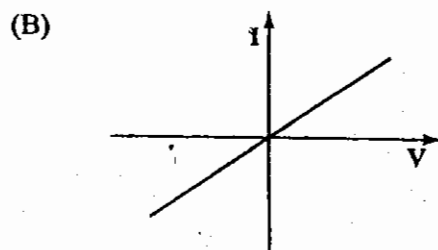
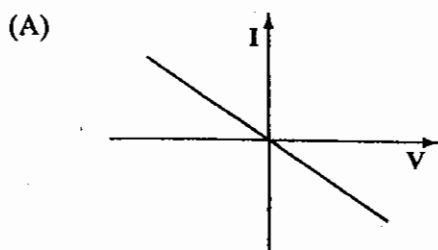
39. Which of the following formulae (in which the symbols have their usual meaning) is a mathematical expression of Ohm's Law?

- (A) $V = IR$
(B) $P = VI$
(C) $Q = IT$
(D) $E = PT$

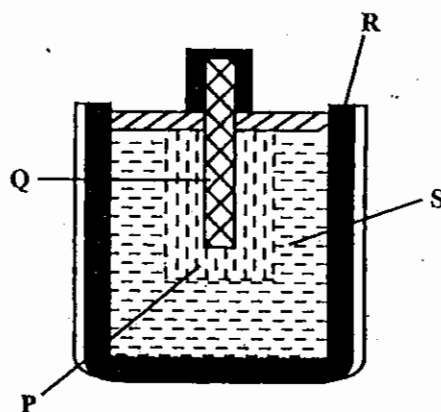
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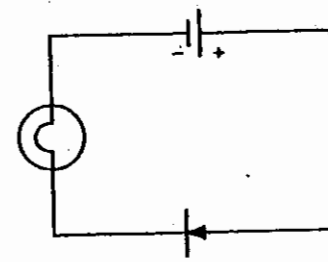
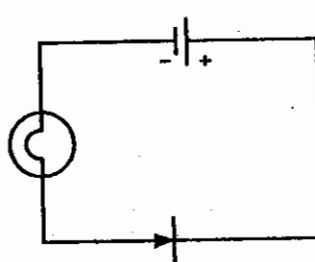
40. Which of the following is a representation of the current versus p.d. relationship for a metallic conductor at a constant temperature?



Item 42 refers to the following diagram.



42. Which of the following is correct?
- (A) Q is carbon, R is zinc.
 - (B) P is copper oxide, Q is carbon.
 - (C) R is iron and S is ammonium chloride.
 - (D) S is sulphuric acid and P is manganese dioxide.



43. A simple experiment was conducted using the circuit diagrams shown above. The same components were used and the bulb was lit to normal brightness in each case.

Which of the following statements would be correct?

- I. The bulb is defective.
- II. The battery is defective.
- III. The diode is defective.

- (A) I only
- (B) III only
- (C) I and II only
- (D) II and III only

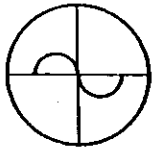
41. In domestic installation systems, which of the following should be placed in the live wire?

- I. Switches
- II. Circuit breakers
- III. Fuses

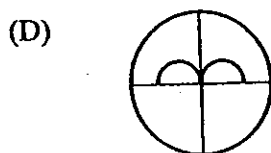
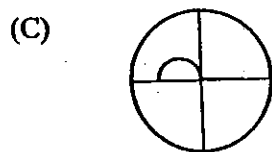
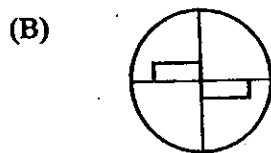
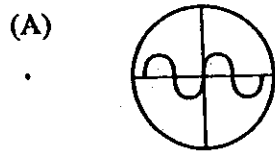
- (A) I only
- (B) III only
- (C) II and III only
- (D) I, II and III

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44. A sinusoidal voltage was displayed as shown on the diagram of an oscilloscope screen below.



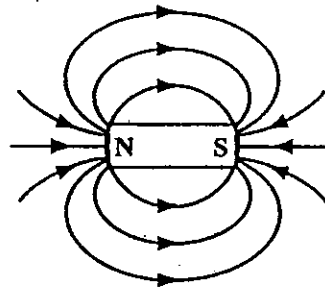
If the voltage was rectified using a SINGLE semiconductor diode, which of the following would represent the trace if the same timebase were applied?



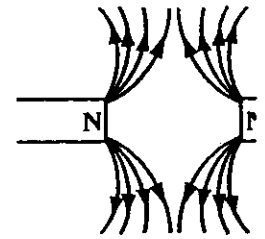
45. Which of the following pairs of statements is true for both iron and steel?

	Iron is	Steel
(A)	easily magnetised	does not retain magnetism
(B)	not easily magnetised	retains its magnetism well
(C)	easily magnetised	retains its magnetism well
(D)	not easily magnetised	does not retain its magnetism

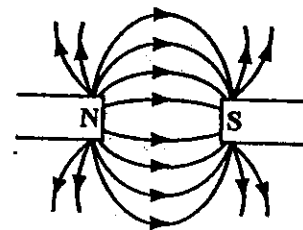
Item 46 refers to the diagrams below showing the magnetic field lines plotted by a student.



I



II



III

46. Which of the following are correct?

- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III only

47. Which of the following is MOST suitable for the core of an electromagnet?

- (A) Copper
(B) Steel
(C) Carbon
(D) Soft iron

48. A glass rod is rubbed with a piece of silk and is positively charged. The glass rod became charged by

- (A) gaining protons
(B) losing protons
(C) gaining electrons
(D) losing electrons

49. Which of the following make use of the magnetic field generated by an electric current?

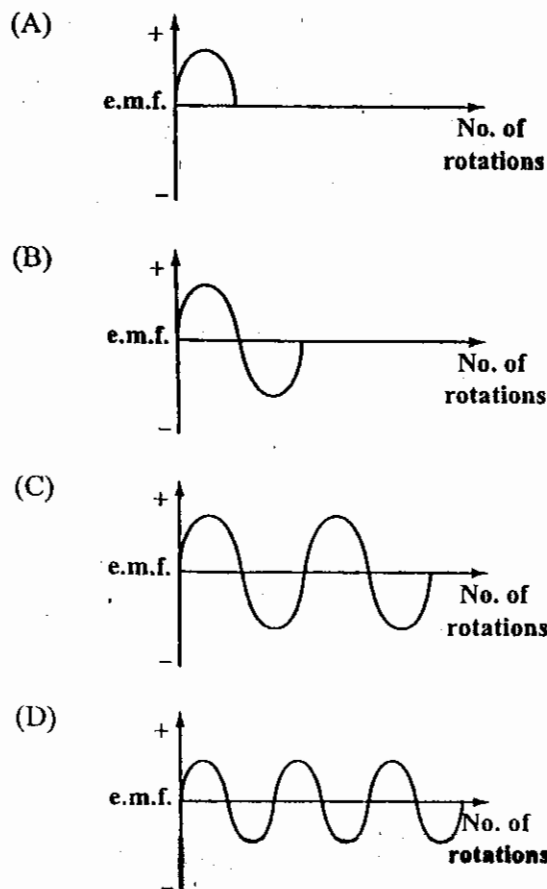
- I. d.c. motor
- II. Moving
- III. Electric iron

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

50. It is possible to induce a voltage in a wire coil by the use of a

- (A) steady magnetic field
- (B) positively charged rod
- (C) negatively charged rod
- (D) changing magnetic field

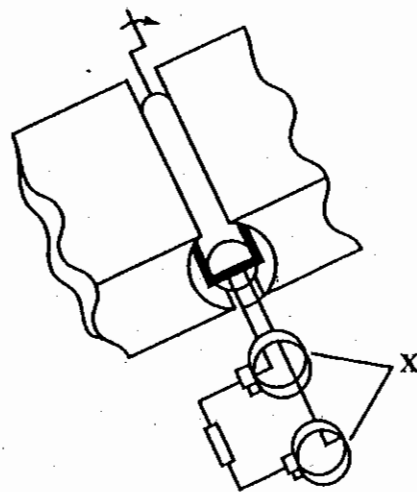
51. For a simple a.c. generator, which of the following graphs will be produced by two complete rotations of the armature?



52. The electric power supplied to homes and to factories is alternating rather than direct because

- (A) the use of a.c. reduces electrical hazards
- (B) the use of a.c. reduces transmission losses
- (C) a.c. enables a wider variety of appliances to be used
- (D) a.c. power lines are less likely to be struck by lightning than d.c. power lines

Item 53 refers to the following diagram of a simple a.c. generator.

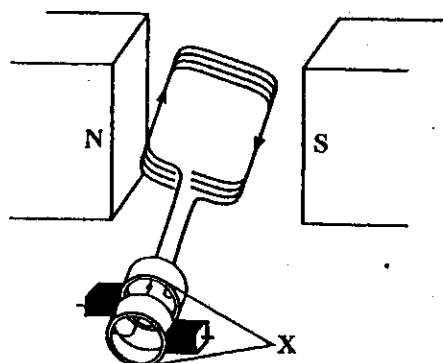


53. The parts labelled X in the diagram are known as the

- (A) coil
- (B) armature
- (C) slip rings
- (D) commutator

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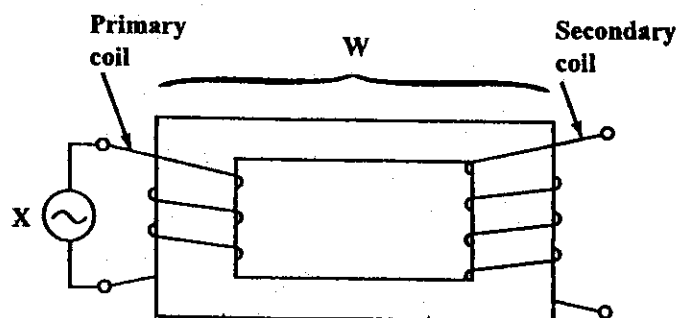
Item 54 refers to the following diagram.



54. Which of the following does the diagram above represent?

- (A) An a.c. generator
- (B) A d.c. generator
- (C) A voltmeter
- (D) An ammeter

Item 55 refers to the following diagram.



55. Appropriate labels for W and X would be

- | | W | X |
|-----|-----------------------|------------|
| (A) | step-down transformer | a.c. input |
| (B) | step-down transformer | d.c. input |
| (C) | step-up transformer | a.c. input |
| (D) | step-up transformer | d.c. input |

56. Which of the following is an isotope of $^{12}_6\text{X}$?

- (A) ^6_3X
- (B) $^{13}_6\text{X}$
- (C) $^{12}_{10}\text{X}$
- (D) ^3_1X

57. Which of the following statements may be used to define the term "half-life"?

- I. It is half the time required by a nucleus to decay.
- II. It is the time for half of the nuclei of a sample to decay.
- III. It is the time for the activity of any random process to fall to one half of its original value.

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

58. When alpha particles, beta particles and gamma rays from a radioactive source are subjected to a transverse magnetic field, which of them would NOT be deviated from their original path?

- I. Alpha
- II. Beta
- III. Gamma

- (A) II only
- (B) III only
- (C) I and II only
- (D) II and III only

59. An isotope of uranium, ${}_{92}^{234}\text{U}$, changes to ${}_{90}^{230}\text{Th}$.

This is an example of

- (A) fission
- (B) fusion
- (C) α -decay
- (D) β -decay

60. In the equation $\Delta E = \Delta mc^2$

- (A) c = speed of light, Δm = mass of atom
- (B) c = specific heat capacity of substance, Δm = mass of substance
- (C) c = specific heat capacity of substance, Δm = mass lost
- (D) c = speed of light, Δm = mass lost

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.